5 Artificial Intelligence

Artificial intelligence is a category of technology that allows computers and machines to emulate human capabilities such as learning and understanding, as well as problem-solving, decision-making, creativity, and autonomous action.

5.1 Agentic AI

Agentic AI refers to autonomous or semi-autonomous AI software systems (AI agents) designed to achieve specific clinical or operational goals with minimal, or no, constant human supervision.

5.1.1 Does the hospital use agentic AI?

- □ Yes, fully implemented (hospital-wide/all departments)
- □ X Yes, partially implemented (selected departments/piloting)
- □ Not yet, but this is planned or in early stages of implementation
- □ No, this is not available

5.1.2 What tool(s) or supplier(s) does the hospital use for agentic AI?

[Pusula PRM (CRM); medguide.app; as well as AI-based applications developed in-house by the hospital, with development ongoing.]

5.1.3 For what tasks is agentic AI used in the hospital? You may select multiple answers.

- □ Clinical decision support (e.g., suggesting diagnoses and treatments, identifying patterns)
- □ X I T support tasks (e.g., first-level support, directing queries)
- □ X Resource management (e.g., bed management, stocking and inventory ordering)
- □ X Patient engagement (e.g., post-discharge follow-up, triaging symptoms)
- □ X Patient intake and discharge (e.g., admission, identity verification)
- □ X Time management (e.g., appointment scheduling, reminders, calendar management)
- □ Other *[Text field]*

5.1.4 What percentage of patient visits use agentic AI?

- □ X Less than 10%
- □ Around 10% to 20%
- □ Around 20% to 50%
- □ More than 50%

5.1.5 Does the hospital perform evaluations of the implementation and efficacy of agentic AI?

- □ X Yes, regularly
- □ Yes, after a new tool is implemented
- □ No

5.1.6 Please provide evidence (e.g., dashboards) of the hospital's implementation of agentic AI.

Please do not upload any evidence containing personally identifiable or sensitive data.

[call center analizi eklenecek]

5.2 Ambient AI

Ambient AI refers to artificial intelligence systems that operate in the background to passively "listen" to and observe clinician-patient interactions, automatically converting them into structured clinical documentation and actionable insights.

5.2.1 Does the hospital use ambient AI?

- ☐ Yes, fully implemented (hospital-wide/all departments)
- ☐ Yes, partially implemented (selected departments/piloting)
- ☐ X Not yet, but this is planned or in early stages of implementation
- ☐ No, this is not available

5.2.2 What tool(s) or supplier(s) does the hospital use for ambient AI?

[]

5.2.3 In which settings does the hospital use ambient AI? You may select multiple answers.

- ☐ Ambulatory services on the hospital site
- ☐ Ambulatory services in offsite/affiliated clinics
- ☐ Inpatient services
- ☐ Other settings *[Text field]*

5.2.4 What percentage of ambulatory patient visits use ambient AI?

- ☐ Less than 10%
- ☐ Around 10% to 20%
- ☐ Around 20% to 50%
- ☐ More than 50%

5.2.5 What percentage of inpatient visits use ambient AI?

- ☐ Less than 10%
- ☐ Around 10% to 20%
- ☐ Around 20% to 50%
- ☐ More than 50%

5.2.6 Does the hospital perform evaluations of the implementation and efficacy of ambient AI?

- ☐ Yes, regularly
- ☐ Yes, after a new tool is implemented
- ☐ No

5.2.7 Please provide evidence (e.g., dashboards) of the hospital's implementation of ambient AI.

Please do not upload any evidence containing personally identifiable or sensitive data.

[Upload field]

5.3 Administrative Support

Administrative support refers to the use of artificial intelligence for tasks relating to administration, such as data entry, scheduling, emails/reminders, or inventory management. This can include agentic AI, ambient AI, or human-prompted tools.

5.3.1 Are AI tools used in the hospital for administrative support tasks?

- ☐ Yes, fully implemented (hospital-wide/all departments)
- ☐ X Yes, partially implemented (selected departments/piloting)
- ☐ Not yet, but this is planned or in early stages of implementation
- ☐ No, this is not available

5.3.2 Please name the AI tools or platforms that are used for administrative support.

[Sci-Kit Learn Library, All Forest Algorithm Models, DeepLearn (Neural Networks), CTrial]

5.3.3 For what administrative support tasks are these AI tools used in the hospital? You may select multiple answers.

- ☐ X Data entry and transcription of results
- ☐ X Financial tasks (e.g., billing & invoicing, salary, reimbursement, insurance)
- ☐ X Logistics (e.g., patient transport, materials requirements planning, inventory)
- ☐ X Patient services (e.g., appointment reminders, admission, discharge letter generation)
- ☐ X Staff time management (e.g., scheduling and shifts, calendar management)
- ☐ Other *[Text field]*

5.3.4 What proportion of administrative work is assisted by AI tools?

- ☐ X Less than 20%
- ☐ Around 20% to 50%
- ☐ Around 50% to 80%
- ☐ More than 80%

5.4 Clinical – Laboratory

5.4.1 Are AI tools used in the hospital for diagnostic or analytic purposes with laboratory data?

- ☐ Yes, fully implemented (hospital-wide/all departments)
- ☐ Yes, partially implemented (selected departments/piloting)
- ☐ X Not yet, but this is planned or in early stages of implementation
- ☐ No, this is not available
- ☐ Other *[Text field]*

5.4.2 Please name the AI tools or platforms that are used for the purpose of diagnosis or analytics using laboratory results.

[Text field]

5.4.3 For which diagnostic or analytic tasks are these AI tools used in the hospital? You may select multiple answers.

- ☐ Assessment of intervention efficacy
- ☐ Automated reporting
- ☐ Chronic illness progression
- ☐ Early warning alerts
- ☐ Identifying infection patterns
- ☐ Prioritization and triage
- ☐ Risk prediction/scoring
- ☐ Quality control
- ☐ Other *[Text field]*

5.4.4 What proportion of clinical laboratory work is assisted by AI tools?

- ☐ Less than 20%
- ☐ Around 20% to 50%
- ☐ Around 50% to 80%
- ☐ More than 80%

5.5 Clinical – Imaging

5.5.1 Are AI tools used in the hospital for diagnostic or analytic purposes with imaging and pathology data?

- ☐ Yes, for analytics only
- ☐ Yes, for diagnosis and analytics
- ☐ X No
- ☐ Other *[Text field]*

5.5.2 Please name the AI tools or platforms that are used for the purpose of diagnosis or analytics using imaging data.

[Text field]

5.5.3 For which diagnostic or analytic tasks are these AI tools used in the hospital? You may select multiple answers.

- ☐ Detecting tumors/foreign bodies
- ☐ Measuring tumor size/volume
- ☐ Detecting hemorrhagic/ischemic stroke
- ☐ Detecting hemorrhage/clots elsewhere in the body
- ☐ Quality control
- ☐ Content-based image retrieval
- ☐ Automated reporting
- ☐ Other *[Text field]*

5.5.4 What proportion of clinical imaging work is assisted by AI tools?

- □ Less than 20%
- □ Around 20% to 50%
- □ Around 50% to 80%
- □ More than 80%

5.6 Clinical – Pathology

5.6.1 Are AI tools used in the hospital for diagnostic or analytic purposes with pathology data?

- □ Yes, for analytics only
- □ Yes, for diagnosis and analytics
- □ X No
- □ Other *[Text field]*

5.6.2 Please name the AI tools or platforms that are used for the purpose of diagnosis or analytics using pathology results.

[Text field]

5.6.3 For which diagnostic or analytic tasks are these AI tools used in the hospital? You may select multiple answers.

- □ Automated reporting
- □ Cell-counting
- □ Detecting tumors/foreign bodies
- □ Measuring tumor size/volume
- □ Image segmentation (e.g., by cell-type)
- □ Quantification of markers (e.g., immunohistochemical staining)
- □ Quality control
- □ Content-based image retrieval
- □ Other *[Text field]*

5.6.4 What proportion of clinical pathology work is assisted by AI tools?

- □ Less than 20%
- □ Around 20% to 50%
- □ Around 50% to 80%
- □ More than 80%

5.7 Patient Support

5.7.1 Are AI tools used in the hospital for patient support tasks?

- □ X Yes, within the hospital only
- □ Yes, remotely (e.g., via an app)
- □ Yes, both within the hospital and remotely
- □ No

5.7.2 Please name the AI tools or platforms that are used for the purpose of patient support.

[Machine Learn Systems, DeepLearn Models, translate.medguide.app/]

5.7.3 For what patient support tasks are these AI tools used in the hospital? You may select multiple answers.

- □ X Patient administration tasks (e.g., first-line enquiries)
- □ X L anguage tasks (e.g., translation, simple language)
- □ Patient education (e.g., symptom management)
- □ Mental health (e.g., anxiety support, digital counselling)
- □ Other

5.7.4 Are AI tools for patient support available via a patient portal?

- □ X Yes
- □ No

5.7.5 What proportion of patients use AI tools for patient support?

- □ X Less than 20%
- □ Around 20% to 50%
- □ Around 50% to 80%
- □ More than 80%

5.8 Other Purposes

5.8.1 Are AI tools used in the hospital for purposes other than those previously mentioned?

- □ X Yes
- □ No

5.8.2 Please name the AI tools and purposes for which these are used in the hospital.

[We use internally developed predictive machine learning models in our call center operations. Our own team builds and maintains these models, which analyze operator efficiency and support decision-making mechanisms aimed at improving operator performance. These tools are already in use, and we are continuing to refine and develop them.]

5.8.3 For what other purposes are these AI tools used in the hospital? You may select multiple answers.

- □ AI-assisted surgery planning
- □ Clinical decision support tools (e.g. therapy recommendations, medication)
- □ Monitoring of clinical parameters (e.g., early detection of sepsis, ventilation monitoring)
- □ Research activities
- □ X Other [We use internally developed predictive machine learning models in our call center operations. Our own team builds and maintains these models, which analyze operator efficiency and support decision-making mechanisms aimed at improving operator performance. These tools are already in use, and we are continuing to refine and develop them.]

5.9 □ AI Tool Support

5.9.1 Is there official support within the hospital for AI tools (e.g., business or premium accounts, dedicated IT team)?

- □ Yes, there is official support
- □ X Yes, there is official support and “power users” (users with additional training)
- □ Yes, there are “power users”
- □ No

5.9.2 For which categories of AI tools is support available? You may select multiple answers.

- □ X Administrative support
- □ Laboratory
- □ Imaging
- □ Pathology
- □ X Patient support
- □ Other *[Text field]*